Department of Mechanical and Production Engineering
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Take-Home Assignment Lec. 14–17

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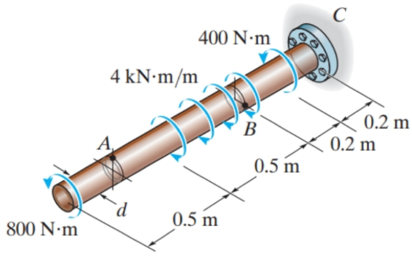
Problem 10-30

The solid shaft is subjected to the distributed and concentrated torsional loadings shown. Determine the required diameter d of the shaft if the allowable shear stress for the material is $\tau_{\text{allow}} = 60\,\text{MPa}$.



Derivation

We let x denote the distance from the free end of the shaft (i.e. $C_x = 1{,}4\,\text{m}$). The shaft is subject to two concentrated torsional loadings, one distributed torsional loading and an unknown reaction torsion in C, which we will denote T_C. Hence, the torque equilibrium for the entire shaft yields:

$$\begin{aligned} 0 &= -800\,\text{N m} - 400\,\text{N m} + 4000\,\text{N m}/\text{m} \cdot 0{,}5\,\text{m} + T_C \\ T_C &= (1200 - 2000)\,\text{N m} \\ &= -800\,\text{N m}. \end{aligned}$$

Therefore the wall will exert a reaction torque of $-800\,\text{N m}$ to the shaft (this is in the same direction as the two concentrated loadings per the sign).

We now split up the shaft into the following four sections:

Section 1	$0 \leq x < 0{,}5\,\text{m}$
Section 2	$0{,}5\,\text{m} \leq x \leq 1{,}0\,\text{m}$
Section 3	$1{,}0\,\text{m} < x \leq 1{,}2\,\text{m}$
Section 4	$1{,}2\,\text{m} < x \leq 1{,}4\,\text{m}$

In section 1, only the $-800\,\text{N m}$ torque acts on the left left side of a cut meaning the internal torque here is $T_1 = 800\,\text{N m}$. In section 2 the distributed torque will have a total magnitude of $4000\,\text{N m}/\text{m} \cdot 0{,}5\,\text{m} = 2000\,\text{N m}$. This means, starting at $x = 0{,}5\,\text{m}$ the internal torque T, will go from $T(0{,}5\,\text{m}) = 800\,\text{N m}$ to $T(1{,}0\,\text{m}) = -800\,\text{N m} - 2000\,\text{N m} = -1200\,\text{N m}$. In section 3, all of the distributed torque has acted and a cut on the left hand side will therefore sill have an internal torque of $T_3 = T(1{,}0\,\text{m}) = -1200\,\text{N m}$. In section 4 an concentrated loading of $-400\,\text{N m}$ is also applied meaning that the internal torque here must become $T_4 = -800\,\text{N m}$.

Therefore the maximum internal torque is in sections 2 and 3 at $-1200\,\text{N m}$. The torsion formula is:

$$\tau_{\max} = \frac{T_{\max} \cdot c}{J}.$$

For a solid circular shaft as here the distance from the center $c = r = d/2$ and the polar moment of inertia $J = \pi d^4/32$. Solving for the required diameter d we get:

$$\begin{aligned} \tau_{\max} &= \frac{T_{\max} \cdot \frac{d}{2}}{\frac{\pi d^4}{32}} \\ &= \frac{16 T_{\max}}{\pi d^3} \\ \implies d &= \sqrt[3]{\frac{16 T_{\max}}{\pi \tau_{\max}}} \\ &= \sqrt[3]{\frac{16 \cdot 1200\,\text{N m}}{\pi \cdot 60\,\text{MPa}}} \end{aligned}$$

$$= 0{,}047\,\mathrm{m} = 47\,\mathrm{mm}.$$

$$\boxed{d = 47\,\mathrm{mm}}$$ RESULT

Problem 10-38

The propeller of a ship is connected to an A–36 steel shaft that is 60 m long and has an outer diameter of 340 mm and an inner diameter of 260 mm. If the power output is 4,5 MW when the shaft rotates at $20\,\mathrm{s}^{-1}$, determine the maximum torsional stress in the shaft and its angle of twist.

Derivation

We start by converting the power output and frequency to an equivalent torque using the power formula:

$$\begin{aligned} P &= T\omega \\ T &= \frac{P}{\omega} \\ &= \frac{4{,}5 \cdot 10^6\,\mathrm{W}}{20\,\mathrm{s}^{-1}} \\ &= 225 \cdot 10^3\,\mathrm{N\,m}. \end{aligned}$$

A tubular shaft has a polar moment of inertia of:

$$J = \frac{\pi}{2}\left(c_o^4 - c_i^4\right) = \frac{\pi}{2}\left((170\,\mathrm{mm})^4 - (130\,\mathrm{mm})^4\right) = 8{,}6 \cdot 10^{-4}\,\mathrm{m}^4.$$

Therefore, using the torsion formula we can find the maximum torsional stress as:

$$\begin{aligned} \tau &= \frac{Tc}{J} \\ &= \frac{225 \cdot 10^3\,\mathrm{N\,m} \cdot 0{,}17\,\mathrm{m}}{8{,}6 \cdot 10^{-4}\,\mathrm{m}^4} \\ &= 44{,}5\,\mathrm{MPa}. \end{aligned}$$

The angle of shift can be found using the formula:

$$\phi = \frac{TL}{JG}.$$

The shear modulus of A–36 steel is $G = 79{,}3\,\mathrm{GPa}$. Inputting all known values we obtain:

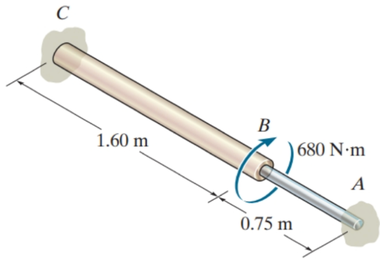
$$\begin{aligned} \phi &= \frac{225 \cdot 10^3\,\mathrm{N\,m} \cdot 60\,\mathrm{m}}{8{,}6 \cdot 10^{-4}\,\mathrm{m}^4 \cdot 79{,}3 \cdot 10^9\,\mathrm{Pa}} \\ &= 0{,}198 = 11{,}3^\circ. \end{aligned}$$

$$\boxed{\begin{aligned} &\tau = 44{,}5\,\mathrm{MPa} \\ &\phi = 0{,}198 = 11{,}3^\circ. \end{aligned}}$$

RESULT

Problem 10-72

A rod is made from two segments: AB is steel and BC is brass. It is fixed at its ends and subjected to a torque of $T = 680\,\mathrm{N\,m}$. If the steel portion has a diameter of 30 mm, determine the required diameter of the brass portion so the reactions at the walls will be the same. Take $G_{st} = 75\,\mathrm{GPa}$, $G_{br} = 39\,\mathrm{GPa}$.



Derivation

Since this is a statically indeterminate problem we must utilize the fact that the rotation at the two ends must be equal as both ends are fixed. Using the formula for the angle of shift and the fact that the angle of shift for both ends must be equal we obtain:

$$\begin{aligned}\phi_{st} &= \phi_{br}\\ \frac{T_{st}L_{st}}{J_{st}G_{st}} &= \frac{T_{br}L_{br}}{J_{br}G_{br}}\\ \frac{L_{st}}{J_{st}G_{st}} &= \frac{L_{br}}{J_{br}G_{br}}\\ J_{br} &= \frac{L_{br}J_{st}G_{st}}{L_{st}G_{br}}.\end{aligned}$$

For a circular solid shaft we have the polar moment of inertia of

$$J = \frac{\pi d^4}{32}.$$

Therefore J_{st} can be found as:

$$J_{st} = \frac{\pi \cdot (0{,}03\,\mathrm{mm})^4}{32} = 7{,}95 \cdot 10^{-8}\,\mathrm{m}^4.$$

Now we can calculate the required polar moment of inertia for the brass segment as:

$$J_{br} = \frac{1{,}60\,\mathrm{m} \cdot 7{,}95 \cdot 10^{-8}\,\mathrm{m}^4 \cdot 75\,\mathrm{GPa}}{0{,}75\,\mathrm{m} \cdot 39\,\mathrm{GPa}} = 3{,}26 \cdot 10^{-7}\,\mathrm{m}^4.$$

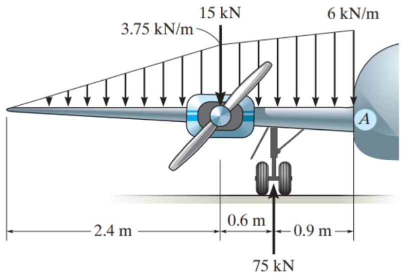
Using the formula for the polar moment of inertia of a solid shaft and solving for the diameter, the required diameter of the brass section can be found as:

$$\begin{aligned}J_{br} &= \frac{\pi d_{br}^4}{32}\\ d_{br} &= \sqrt[4]{\frac{32 J_{br}}{\pi}}\\ &= \sqrt[4]{\frac{32 \cdot 3{,}26 \cdot 10^{-7}\,\mathrm{m}^4}{\pi}}\\ &= 0{,}0427\,\mathrm{m} \approx 43\,\mathrm{mm}.\end{aligned}$$

$$\boxed{d_{br} = 43\,\mathrm{mm}}$$ RESULT

Problem 11-1

The dead-weight loading along the centerline of the airplane wing is shown. If the wing is fixed to the fuselage at A, determine the reactions at A, and then draw the shear and moment diagram for the wing.



Derivation

We start by taking a force equilibrium in the y-direction:

$$\sum F_y = 0$$

$$0 = -3{,}75 \cdot 10^3 \,\frac{\text{N}}{\text{m}} \cdot 1{,}2\,\text{m} \cdot \frac{1}{2} - 15 \cdot 10^3\,\text{N} - \left(1{,}5\,\text{m} \cdot \frac{1}{2} \cdot \left(6 \cdot 10^3 \,\frac{\text{N}}{\text{m}} - 3{,}75 \cdot 10^3 \,\frac{\text{N}}{\text{m}}\right)\right) + 75\,\text{kN} + A_y$$

$$A_y = 3{,}75 \cdot 10^3 \,\frac{\text{N}}{\text{m}} \cdot 0{,}6\,\text{m} + 15 \cdot 10^3\,\text{N} + 0{,}75\,\text{m} \cdot 2{,}25 \cdot 10^3 \,\frac{\text{N}}{\text{m}} - 75\,\text{kN}$$

$$= 56{,}1 \cdot 10^3\,\text{N}.$$

No forces act in the x-direction and hence $\sum F_x = 0 \implies A_x = 0$. A moment equilibrium about A gives:

$$\sum M_A = 0$$

$$0 = 3{,}75 \cdot 10^3 \,\frac{\text{N}}{\text{m}} \cdot 0{,}6\,\text{m} \cdot 2{,}7\,\text{m} + 15 \cdot 10^3\,\text{N} \cdot 1{,}5\,\text{m} + 0{,}75\,\text{m} \cdot 2{,}25 \cdot 10^3 \,\frac{\text{N}}{\text{m}} \cdot 0{,}75\,\text{m} - 75\,\text{kN} \cdot 0{,}9\,\text{m} + M_A$$

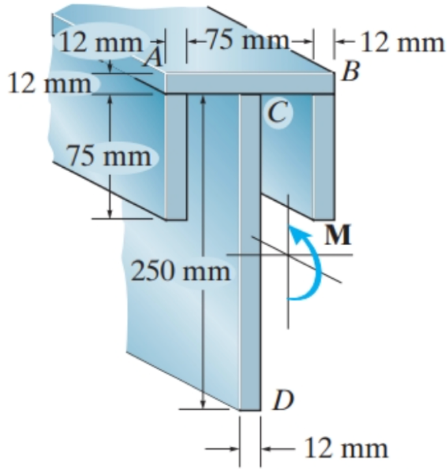
$$M_A = 75\,\text{kN} \cdot 0{,}9\,\text{m} - \left(3{,}75 \cdot 10^3 \,\frac{\text{N}}{\text{m}} \cdot 0{,}6\,\text{m} \cdot 2{,}7\,\text{m} + 15 \cdot 10^3\,\text{N} \cdot 1{,}5\,\text{m} + 0{,}75\,\text{m} \cdot 2{,}25 \cdot 10^3 \,\frac{\text{N}}{\text{m}} \cdot 0{,}75\,\text{m}\right)$$

$$= 37{,}7 \cdot 10^3\,\text{N}\,\text{m}.$$

$$\boxed{A_x = 0, \quad A_y = 56{,}1 \cdot 10^3\,\text{N}, \quad M_A = 37{,}7 \cdot 10^3\,\text{N}\,\text{m}}$$ RESULT

Problem 11-42

Determine the maximum tensile and compressive bending stress in the beam if it is subjected to a moment of $M = 6\,\mathrm{kN\,m}$.



Derivation

We consider the top of the beam as our origin and measure y downwards. To find the neutral axis $\overline{y}$ we will use the formula:

$$\overline{y} = \frac{\sum y_i A_i}{\sum A_i}.$$

We break the beam into 4 sections as shown on the figure from the problem statement. Doing this we get

$$\begin{aligned}\overline{y} &= \frac{99\,\mathrm{mm}\cdot 12\,\mathrm{mm}\cdot 6\,\mathrm{mm} + 2\cdot(12\,\mathrm{mm}\cdot 75\,\mathrm{mm}\cdot 43{,}5\,\mathrm{mm}) + 12\,\mathrm{mm}\cdot 250\,\mathrm{mm}\cdot 131\,\mathrm{mm}}{99\,\mathrm{mm}\cdot 12\,\mathrm{mm} + 75\,\mathrm{mm}\cdot 12\,\mathrm{mm} + 12\,\mathrm{mm}\cdot 250\,\mathrm{mm}} \\ &= \frac{99\,\mathrm{mm}\cdot 6\,\mathrm{mm} + 2\cdot 75\,\mathrm{mm}\cdot 43{,}5\,\mathrm{mm} + 250\,\mathrm{mm}\cdot 131\,\mathrm{mm}}{99\,\mathrm{mm} + 75\,\mathrm{mm} + 250\,\mathrm{mm}} \\ &= 94{,}0\,\mathrm{mm}.\end{aligned}$$

Therefore, the neutral axis is 94,0 mm below the top surface. Now we must find the moment of inertia about this axis. We will use the parallel-axis theorem for each rectangular area as:

$$I_{NA} = \sum\left(I_{c,i} + A_i d_i^2\right).$$

Each of the rectangles have an "initial" moment of inertia of $1/12bh^3$. Using this we can calculate the "initial" moment of inertia of each rectangular "plate" as:

$$\begin{aligned}I_{c,i,\mathrm{top}} &= \frac{1}{12}\cdot 99\,\mathrm{mm}\cdot(12\,\mathrm{mm})^3 = 14{,}3\cdot 10^3\,\mathrm{mm}^4 \\ I_{c,i,\mathrm{side}} &= \frac{1}{12}\cdot 12\,\mathrm{mm}\cdot(75\,\mathrm{mm})^3 = 422\cdot 10^3\,\mathrm{mm}^4 \\ I_{c,i,\mathrm{bottom}} &= \frac{1}{12}\cdot 12\,\mathrm{mm}\,(250\,\mathrm{mm})^3 = 15{,}6\cdot 10^6\,\mathrm{mm}^4.\end{aligned}$$

Now, we will find the "shifting" term for each rectangle as:

$$\begin{aligned}\left(A_i d_i^2\right)_{\mathrm{top}} &= 12\,\mathrm{mm}\cdot 99\,\mathrm{mm}\,(94{,}0\,\mathrm{mm} - 6\,\mathrm{mm})^2 = 9{,}21\cdot 10^6\,\mathrm{mm}^4 \\ \left(A_i d_i^2\right)_{\mathrm{side}} &= 12\,\mathrm{mm}\cdot 75\,\mathrm{mm}\cdot(94{,}0\,\mathrm{mm} - 43{,}5\,\mathrm{mm})^2 = 2{,}30\cdot 10^6\,\mathrm{mm}^4 \\ \left(A_i d_i^2\right)_{\mathrm{bottom}} &= 12\,\mathrm{mm}\cdot 250\,\mathrm{mm}\,(94{,}0\,\mathrm{mm} - 131\,\mathrm{mm})^2 = 4{,}11\cdot 10^6\,\mathrm{mm}^4 \\ &.\end{aligned}$$

Now the total moment of inertia about the neutral axis can be calculated as:

$$\begin{aligned} I_{NA} &= 14{,}3\cdot 10^3\,\text{mm}^4 + 422\cdot 10^3\,\text{mm}^4 + 15{,}6\cdot 10^6\,\text{mm}^4 + 9{,}21\cdot 10^6\,\text{mm}^4 + 2{,}30\cdot 10^6\,\text{mm}^4 + 4{,}11\cdot 10^6\,\text{mm}^4 \\ &= 31{,}7\cdot 10^6\,\text{mm}^4. \end{aligned}$$

The flexure formula is

$$\sigma = \frac{Mc}{I}.$$

Where c is the distance to the outer fiber. The maximum compressive bending stress will be:

$$(\sigma_c)_{\max} = \frac{6\,\text{kN m}\cdot 94{,}0\,\text{mm}}{31{,}7\cdot 10^6\,\text{mm}^4} = 17{,}8\,\text{MPa}.$$

And the maximum tensile bending stress will be:

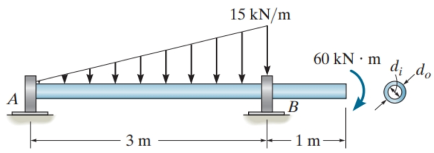
$$(\sigma_t)_{\max} = \frac{6\,\text{kN m}\cdot(262\,\text{mm} - 94{,}0\,\text{mm})}{31{,}7\cdot 10^6\,\text{mm}^4} = 31{,}7\,\text{MPa}.$$

$$\boxed{\begin{aligned} (\sigma_c)_{\max} &= 17{,}8\,\text{MPa} \\ (\sigma_t)_{\max} &= 31{,}7\,\text{MPa}. \end{aligned}}$$

RESULT

Problem 11-70

The tubular shaft is to have a cross section such that its inner diameter and outer diameter are related by $d_i = 0{,}8 d_0$. Determine these required dimensions if the allowable bending stress is $\sigma_{\text{allow}} = 155\,\text{MPa}$.



Derivation

First we must find the point at which the largest internal torque is reached. We start by utilizing a force balance in the y direction, as:

$$\begin{aligned}\sum F_y &= 0\\ R_A - \frac{1}{2}\cdot 15\,\frac{\text{kN}}{\text{m}}\cdot 3\,\text{m} + R_B &= 0\\ \implies R_B &= \frac{1}{2}\cdot 15\,\frac{\text{kN}}{\text{m}}\cdot 3\,\text{m} - R_A. \end{aligned} \tag{0.1}$$

Taking a moment equilibrium about A gives:

$$\begin{aligned}0 &= \sum M_A\\ 0 &= R_B\cdot 3\,\text{m} - \frac{2}{3}\cdot 3\,\text{m}\cdot\left(\frac{1}{2}\cdot 15\,\frac{\text{kN}}{\text{m}}\cdot 3\,\text{m}\right) - 60\,\text{kN}\,\text{m}\\ \left(\frac{1}{2}\cdot 15\,\frac{\text{kN}}{\text{m}}\cdot 3\,\text{m} - R_A\right)\cdot 3\,\text{m} &= \frac{2}{3}\cdot 3\,\text{m}\cdot\left(\frac{1}{2}\cdot 15\,\frac{\text{kN}}{\text{m}}\cdot 3\,\text{m}\right) + 60\,\text{kN}\,\text{m}\\ 67{,}5\,\text{kN}\,\text{m} - R_A\cdot 3\,\text{m} &= 105\,\text{kN}\,\text{m}\\ R_A &= \frac{(67{,}5\,\text{kN}\,\text{m} - 105\,\text{kN}\,\text{m})}{3\,\text{m}}\\ &= -12{,}5\,\text{kN}.\end{aligned}$$

Now inserting this into Equation (0.1) we get:

$$\begin{aligned}R_B &= \frac{1}{2}\cdot 15\,\frac{\text{kN}}{\text{m}}\cdot 3\,\text{m} + 12{,}5\,\text{kN}\\ &= 35\,\text{kN}.\end{aligned}$$

The moment equilibrium for $0 \le x \le 3\,\text{m}$ is:

$$\begin{aligned}0 &= \sum M_{0\,\text{m}\le x\le 3\,\text{m}}\\ 0 &= -R_A\cdot x + \frac{1}{3}x\left(\frac{1}{2}\cdot 15\,\frac{\text{kN}}{\text{m}}\cdot\frac{x}{3\,\text{m}}x\right) + M\\ M_{0\,\text{m}\le x\le 3\,\text{m}} &= -12{,}5\,\text{kN}\cdot x - \frac{1}{18\,\text{m}}\cdot 15\,\frac{\text{kN}}{\text{m}}\cdot x^3\\ &= -12{,}5\,\text{kN}\cdot x - 0{,}833\,\frac{\text{kN}}{\text{m}^2}\cdot x^3.\end{aligned}$$

To find the largest moment in this interval we set the derivative to zero and get the solutions:

$$x_{\max,1} = 0 \quad \text{and} \quad x_{\max,2} = 1{,}23\,\text{m}.$$

These extrema have moments of:

$$\begin{aligned}M(0\,\text{m}) &= -12{,}5\,\text{kN}\cdot 0\,\text{m} - 0{,}833\,\frac{\text{kN}}{\text{m}^2}\cdot 0\,\text{m}^3 = 0\,\text{kN}\,\text{m}\\ M(1{,}23\,\text{m}) &= 12{,}5\,\text{kN}\cdot 1{,}23\,\text{m} - 0{,}833\,\frac{\text{kN}}{\text{m}^2}\cdot(1{,}23\,\text{m})^3 = 30{,}6\,\text{kN}\,\text{m}.\end{aligned}$$

For $3\,\mathrm{m} \le x \le 4\,\mathrm{m}$ the moment equilibrium is:

$$\begin{aligned}
0 &= \sum M_{3\,\mathrm{m}\le x\le 4\,\mathrm{m}} \\
0 &= -R_A x + \left(\frac{1}{3}\cdot 3\,\mathrm{m} + x - 3\,\mathrm{m}\right)\left(\frac{1}{2}\cdot 15\,\frac{\mathrm{kN}}{\mathrm{m}}\cdot 3\,\mathrm{m}\right) + R_B\,(x - 3\,\mathrm{m}) + M \\
M_{3\,\mathrm{m}\le x\le 4\,\mathrm{m}} &= -12{,}5\,\mathrm{kN}\cdot x - \left(\frac{1}{3}\cdot 3\,\mathrm{m} + x - 3\,\mathrm{m}\right)\left(\frac{1}{2}\cdot 15\,\frac{\mathrm{kN}}{\mathrm{m}}\cdot 3\,\mathrm{m}\right) + 35\,\mathrm{kN}\cdot(x - 3\,\mathrm{m}) \\
&= -60\,\mathrm{kN\,m}.
\end{aligned}$$

The moment is constant $M_{3\,\mathrm{m}\le x\le 4\,\mathrm{m}} = -60\,\mathrm{kN\,m}$ for $3\,\mathrm{m} \le x \le 4\,\mathrm{m}$. This is also the largest internal moment in the shaft.

We know that the polar moment of inertia of a hollow shaft is:

$$I = \frac{\pi}{64}\left(d_i^4 - d_o^4\right).$$

Using the condition that $d_i = 0{,}8 d_0$ we get:

$$I = \frac{\pi}{64}\left(d_i^4 - (0{,}8 d_i)^4\right).$$

Therefore the maximum bending moment is:

$$\begin{aligned}
\sigma_{\max} &= \frac{Mc}{I} \\
&= \frac{60\,\mathrm{kN\,m}\cdot \frac{d_i}{2}}{\frac{\pi}{64}\left(d_i^4 - (0{,}8 d_i)^4\right)}.
\end{aligned}$$

Solving for the diameter we get:

$$\begin{aligned}
d_i &= \sqrt[3]{\frac{64\cdot 60\,\mathrm{kN\,m}}{\pi\cdot 1{,}18\cdot 155\,\mathrm{MPa}}} = 188\,\mathrm{mm} \\
d_o &= 0{,}8\cdot 188\,\mathrm{mm} = 151\,\mathrm{mm}.
\end{aligned}$$

As this is the diameter where the bending stress is below the allowable this must correspond to the smallest allowable measurements.

$$\boxed{\begin{aligned} d_i &> 188\,\mathrm{mm} \\ d_o &> 151\,\mathrm{mm}. \end{aligned}}$$

RESULT